

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #4

Instructions

- TA in charge: Sihui Shen (sihui.shen@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Please refer to myCourses.
- You must electronically submit the assignment in the form of a PDF file (only this format will be accepted) on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	Aneurin Hullah
First student ID number:	261283866
Second student full name:	Alexandre Pujal
Second student ID number:	261183554
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (10%)	Question 2 (25%)	Question 3 (10%)	Question 4 (05%)	Total
Question 1 (12%)	Question 2 (13%)	Question 3 (12%)	Question 4 (13%)	

Paper and Paper Assignment #4

Alexandre PUSA

ID: 261183554

Question 1

$$h(x_1, x_2, x_3) = \sum m(1, 3, 4, 5, 6)$$

$$\text{for } (x_1, x_2) = (0, 0): h(0, 0, x_3) = x_3$$

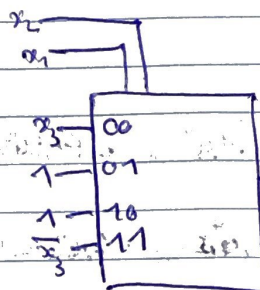
$$(x_1, x_2) = (0, 1): h(0, 1, x_3) = 1$$

$$(x_1, x_2) = (1, 0): h(1, 0, x_3) = 1$$

$$(x_1, x_2) = (1, 1): h(1, 1, x_3) = \overline{x_3}$$

	x_1	x_2	x_3	h
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

Implementation:



Question 2

Inputs: $X: x_3, x_2, x_1, x_0$

$Y: y_3, y_2, y_1, y_0$

$S: s_1, s_0$

Output: Z

$s_1 s_0$	Functionality
00	$Z = X + Y$
01	$Z = X \cdot Y$
10	$Z = X - Y$
11	$Z = 0.5 \cdot X - Y$

In 2's complement, repeating the MSB keeps the sign and magnitude the same (for positive numbers, we extend with 0, and for negative, we extend with 1)

Hence, to avoid overflow, (and to avoid losing the sign when performing the left)

$$X = x_3 x_2 x_1 x_0$$

$$Y = y_3 y_2 y_1 y_0$$

Moreover, $2Y$ is obtained by shifting Y to the left

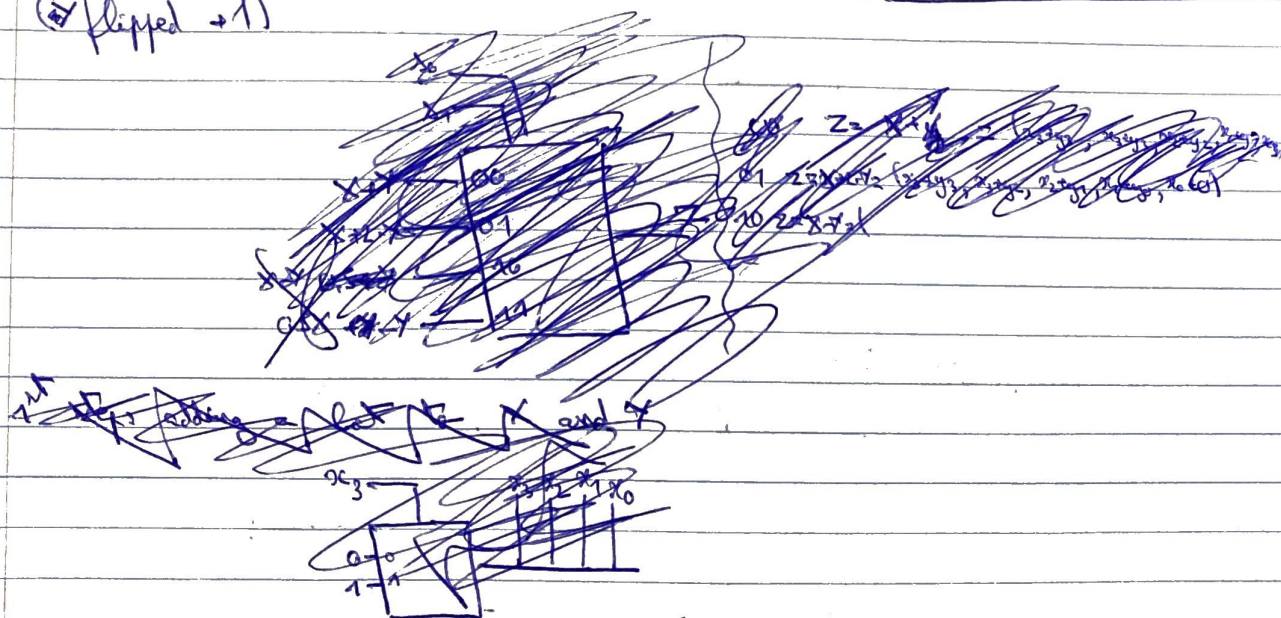
$$\Rightarrow 2Y = y_3 y_2 y_1 y_0 0$$

and $0.5X$ is obtained by shifting X to the right

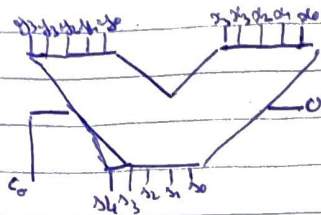
$$\Rightarrow 0.5X = x_3 x_2 x_1 x_0$$

$-Y$ is only the 2's complement of Y : $\neg Y + 1$

(flipped + 1)



$$X+Y:$$

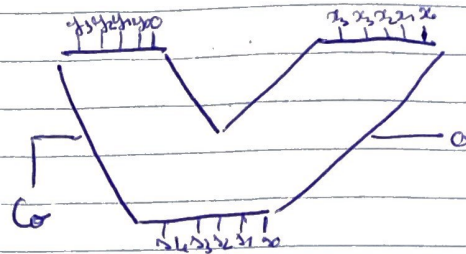


$$2Y: \text{left } Y \text{ to the left}$$

$$Y_3Y_2Y_1Y_0 \rightarrow Y_4Y_3Y_2Y_1$$

$$2Y: Y_4Y_3Y_2Y_1Y_0 = 0$$

$$X+2Y:$$

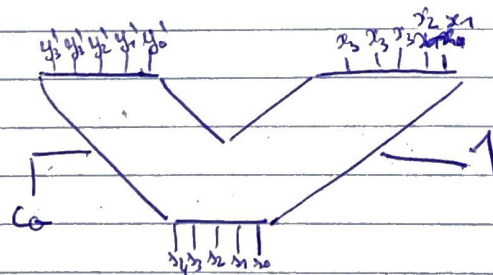


$$O.S.X: \text{left } X \text{ to the right}$$

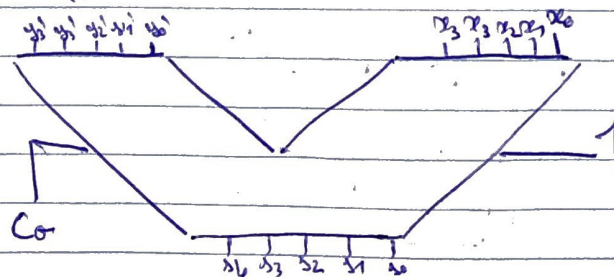
$$X: x_3x_2x_1x_0$$

$$O.S.X: x_3x_2x_1x_0x_1$$

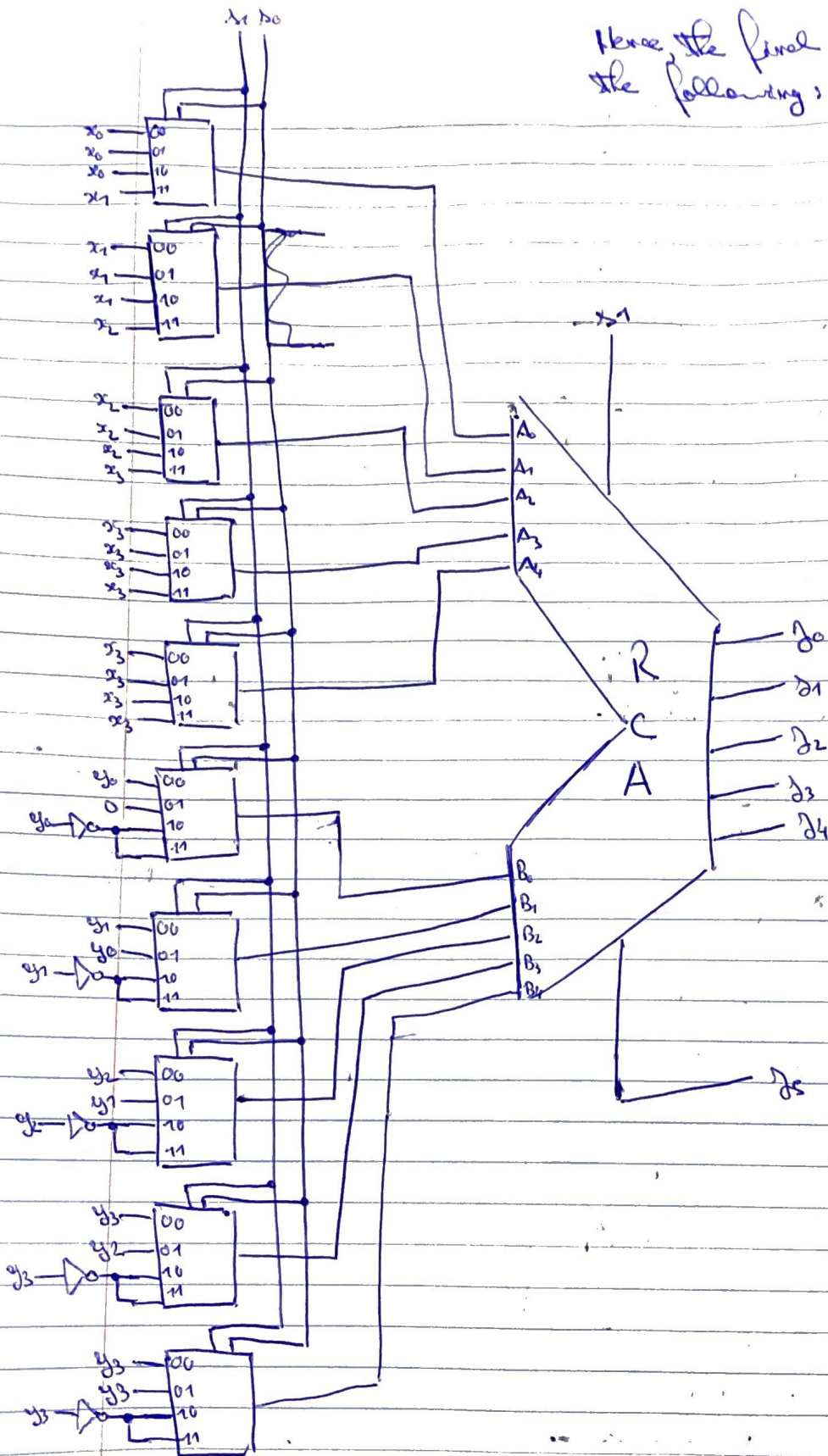
and since in 2's complement: $O.S.X - Y = O.S. + (\bar{Y} + 1)$
 thus, $O.S.X - Y:$



$$\text{and } X - Y = X + (\bar{Y} + 1)$$



Here the final implementation is the following:



Question 3:

a) Full adder

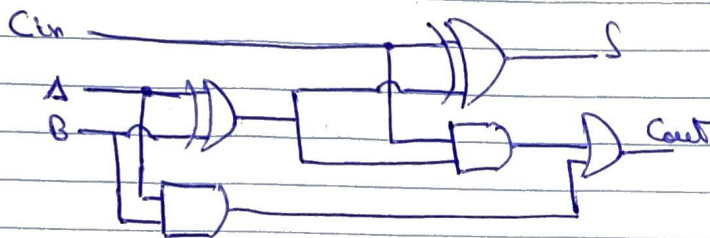
	A	B	Cin	S	cout
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + BC_{in} + AC_{in}$$

4x4 Karnaugh Map for Cout:

AB \ Cin	00	01	11	10
0	0	0	0	0
1	0	1	1	1



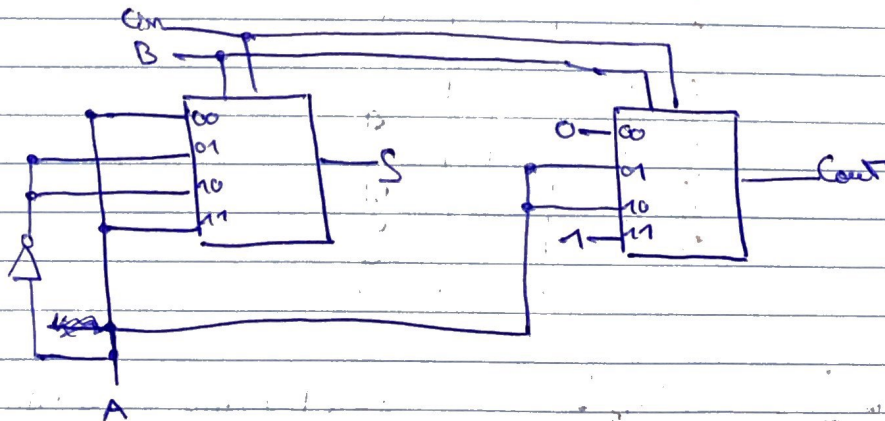
When B Cin = (0,0): S = A Cout = 0

(0,1): S = $\bar{A}$ Cout = A

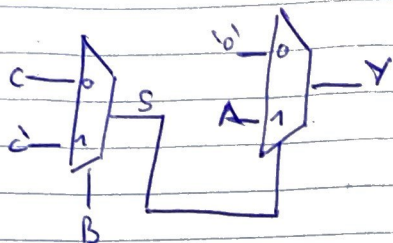
(1,0): S = $\bar{A}$ Cout = A

(1,1): S = A Cout = 1

Hence, we take B and Cin as selection lines



2)



We denote S the output of the first 2-to-1 MUX

$$\begin{aligned}
 Y &= S' \cdot 0 + S \cdot A & \text{with } S &= B'C + BC' \\
 &= (B+C')(B'+C)A + B'CA + BC'A \\
 &= B'CA + BC'A \\
 &= A(B'C + BC') \\
 &= A(B \oplus C)
 \end{aligned}$$

$$\begin{aligned}
 S' &= (B'C + BC')' \\
 &= (B'C)'(BC')' \\
 &= (B+C')(B'+C)
 \end{aligned}$$

Question 4:

Truth Table

	A	B	C	D	Y
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	1
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	0
15	1	1	1	1	1

AB	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	1	1	1	1
10	0	1	0	0

$$Y = CD + \overline{A}BD + \overline{A}BC + AB'D$$

Question 5 (12%)

Implement the given function:

$$f(a, b, c, d, e) = abce + b\bar{d}\bar{e} + \bar{b}de + \bar{b}\bar{e}(a \oplus c) + \bar{a}b\bar{c}e$$

Use a single 4-to-1 multiplexer and as few other gates as possible. Assume that only the uncomplemented form of the inputs is available.

Hint: use Shannon's expansion with respect to variables that appear more often in the function.



Question 6 (13%)

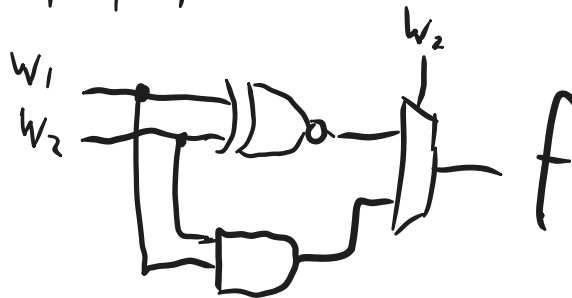
Use Shannon's expansion to implement the function $f(w_1, w_2, w_3) = \sum m(0, 5, 7)$ using 2-to-1 multiplexer and required gates.

	w_1	w_2	w_3	f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

0 0 0
1 0 1
1 1 1

$$\overline{w_1} \overline{w_2} \overline{w_3} + w_1 \overline{w_2} w_3 + w_1 w_2 w_3$$

Shannon's expansion about w_2

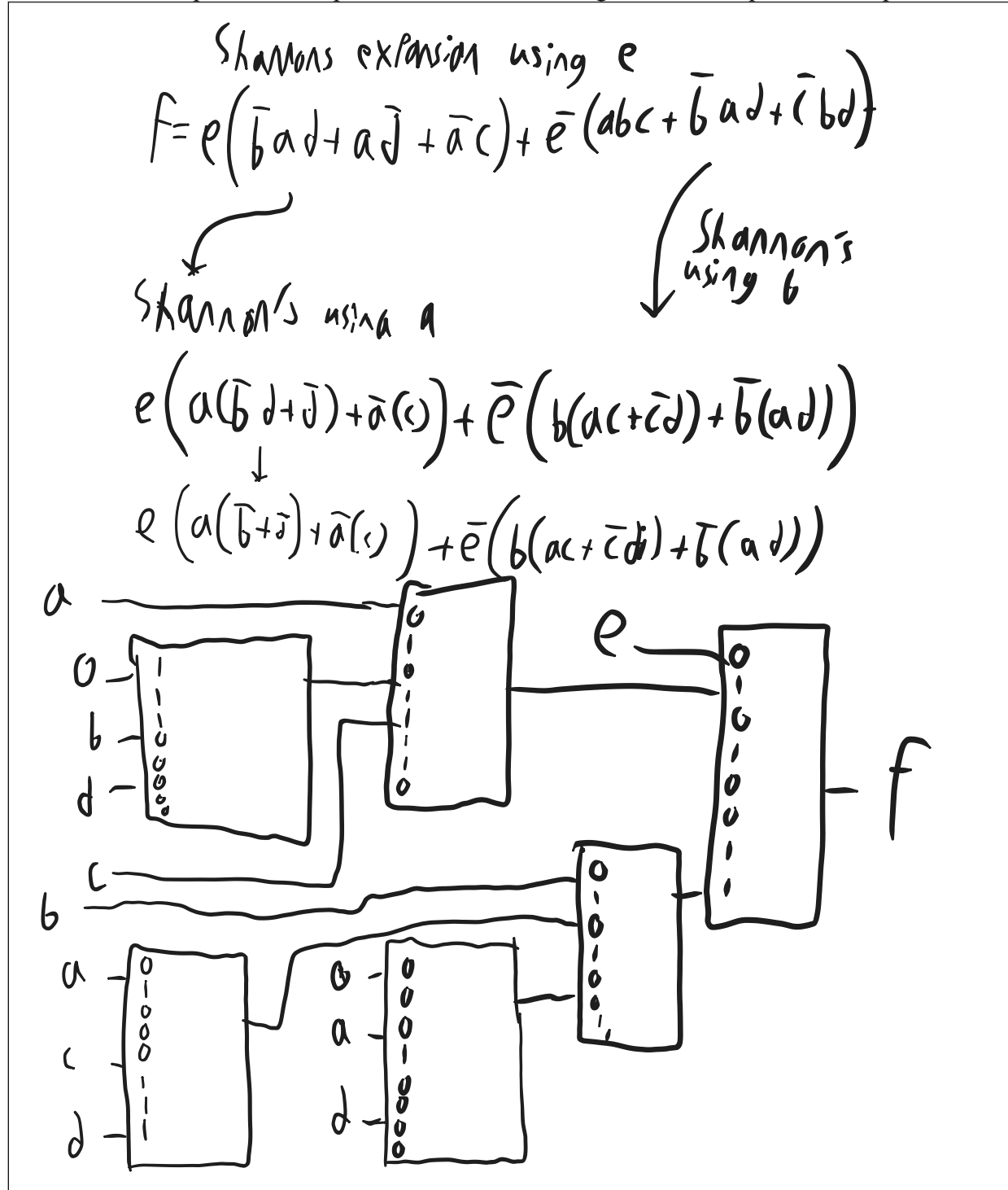


Question 7 (12%)

Given the following function:

$$f(a, b, c, d, e) = abc\bar{e} + \bar{b}ad + \bar{c}\bar{e}bd + a\bar{d}e + \bar{a}ce$$

Use Shannon's expansion to implement the function using as fewer 3-input LUTs as possible.



Question 8 (13%)

Design a comparator circuit that compares two 4-bit unsigned integers A and B. The circuit has three outputs: E equals '1' if $A = B$, G equals '1' if $A > B$ and is '0' otherwise, and L equals '1' if $A < B$ and is '0' otherwise. You can only use XOR and NOT gates, a priority encoder, and exactly two (as small as possible) multiplexers. Provide the circuit diagram of the comparator.

